EXIST 2026

sEXism Identification in Social neTworks

Lab Guidelines

Jorge Carrillo-de-Albornoz[1], Laura Plaza[1], Iván Arcos[2], Maria Aloy Mayo[2], Paolo Rosso[2], Damiano Spina[3]

Source: unsplash

https://nlp.uned.es/exist2026/

[1] Universidad Nacional de Educación a Distancia
[2] Universitat Politècnica de València
[3] RMIT University

Task Description

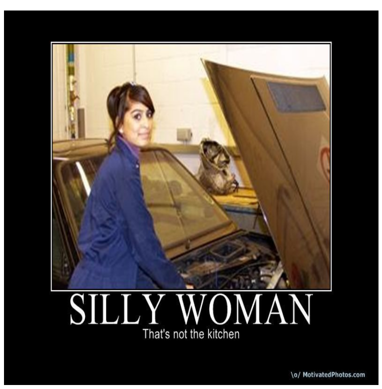
Building upon the **EXIST 2025 dataset**, this edition focuses exclusively on **multimedia formats**, comprising six experimental subtasks applied to **images (memes)** and **videos (TikToks)**. Participants are challenged to address three main objectives: **sexism identification (x.1)**, **source intention detection (x.2)**, and **sexism categorization (x.3)**.

A groundbreaking feature of this lab is the integration of **Human-Centered AI principles**. **In the new experimental framework introduced in EXIST 2026, selected subjects were exposed to the multimedia content while their physiological and behavioral responses were continuously recorded.** These **multimodal signals** (including **eye tracking, heart rate, and EEG**) enrich the traditional annotation labels, providing a deeper window into how users unconsciously process and react to sexist content in **English and Spanish**.

Although the task numbering may seem unusual, it has been kept this way to ensure consistency with the structure of the 2025 dataset and the organization of the EvALL repository.

SUBTASK 2.1: Sexism Identification in Memes

This is a binary classification subtask consisting of deciding whether or not a given meme is sexist, contains sexist expressions or behaviours (i.e., it is sexist itself, describes a sexist situation or criticizes a sexist behaviour), and classify it according to two categories: **YES** and **NO**. The following figures are some examples:



(a) YES

(b) NO

SUBTASK 2.2: Source Intention in Memes

Once a message has been classified as sexist, the second subtask aims to categorize the meme according to the intention of the author, which provides insights in the role played by social networks on the emission and dissemination of sexist messages. Due to the characteristics of the memes, systems should only classify memes with **DIRECT** or **JUDGEMENTAL** labels.

- **DIRECT**: the intention was to write a message that is sexist by itself or incites to be sexist.

- **JUDGEMENTAL**: the intention was to judge, since the tweet describes sexist situations or behaviours with the aim of condemning them.

The following figures are some examples of them, respectively.

(a) Direct

(b) Judgemental

SUBTASK 2.3: Sexism Categorization in Memes
Many facets of a woman's life may be the focus of sexist attitudes including domestic and parenting roles, career opportunities, sexual image, and life expectations, to name a few. Automatically detecting which of these facets of women are being more frequently attacked in social networks will facilitate the development of policies to fight against sexism. According to this, each sexist meme must be categorized in one or more of the following categories

- **IDEOLOGICAL AND INEQUALITY**: The text discredits the feminist movement, rejects inequality between men and women, or presents men as victims of gender-based oppression.
- **STEREOTYPING AND DOMINANCE**: The text expresses false ideas about women that suggest they are more suitable to fulfill certain roles (mother, wife, family caregiver, faithful, tender, loving, submissive, etc.), or inappropriate for certain tasks (driving, hardwork, etc), or claims that men are somehow superior to women.
- **OBJECTIFICATION**: The text presents women as objects apart from their dignity and personal aspects, or assumes or describes certain physical qualities that women must have in order to fulfill traditional gender roles (compliance with beauty standards, hypersexualization of female attributes, women's bodies at the disposal of men, etc.).
- **SEXUAL VIOLENCE**: Sexual suggestions, requests for sexual favors or harassment of a sexual nature (rape or sexual assault) are made.
- **MISOGYNY AND NON-SEXUAL VIOLENCE**: The text expressses hatred and violence towards women.

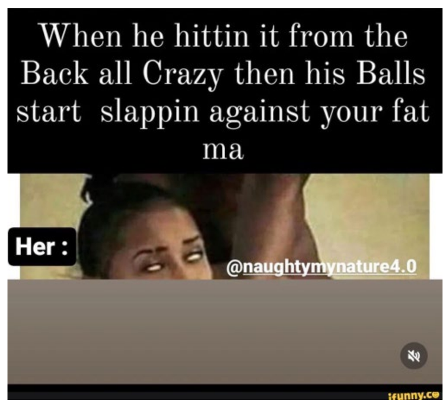
The following figures are some examples of categorized memes.

(a) Stereotyping

(b) Ideological

(c) Objectification

(d) Misogyny



(e) Sexual violence

SUBTASK 3.1: Sexism Identification in Videos
This subtask is the same subtask as subtask 2.1. Some examples of videos classified as YES or NO can be accessed on the website: https://nlp.uned.es/exist2026/

SUBTASK 3.2: Source Intention in Videos
This subtask replicates subtask 2.2 for memes, but it takes as source videos. Some examples of videos classified as DIRECT or JUDGEMENTAL can be accessed on the website: https://nlp.uned.es/exist2026/

SUBTASK 3.3: Sexism Categorization in Videos
This subtask aims to classify sexist videos according to the categorization provided for Task 1.3: (i) IDEOLOGICAL AND INEQUALITY, (ii) STEREOTYPING AND DOMINANCE, (iii) OBJECTIFICATION, (iv) SEXUAL VIOLENCE and (v) MISOGYNY AND NON-SEXUAL

VIOLENCE. Some examples of videos classified can be accessed on the website: https://nlp.uned.es/exist2026/

More details can be found at the EXIST 2026 website (https://nlp.uned.es/exist2026/).

Datasets Description

The EXIST 2026 Memes Dataset will be used in Task 2.X (subtasks 1–3), while the EXIST 2026 Videos Dataset will be used in Task 3.X (subtasks 1–3).

This edition builds upon the EXIST 2025 dataset and extends it with additional sensor-based information collected under a Human-Centered AI framework, enriching the original annotations with complementary physiological data.

EXIST 2026 Memes Dataset

The EXIST 2026 Memes Dataset contains more than 5,000 labeled memes, both in English and Spanish. In particular, the training set contains 3,984 memes and the test set contains 1,053 memes. Distribution between both languages has been balanced. Additionally, a small number of memes from the 2025 edition have been removed after detecting duplicated instances that were not identified during the initial dataset creation process.

The data sets are provided in **JSON format**. Each meme is represented as a JSON object with the following attributes:

1. "**id_EXIST**": a unique identifier for the meme.
2. "**lang**": the languages of the meme ("en" or "es").
3. "**text**": the text automatically extracted from the meme.
4. "**meme**": the name of the file that contains the meme.
5. "**path_memes**": the path to the file that contains the meme.
6. **"number_annotators":** the number of persons that have annotated the meme.
7. **"annotators":** a unique identifier for each of the annotators.
8. **"gender_annotators":** the gender of the different annotators. Possible values are: "F" and "M", for female and male respectively.
9. **"age_annotators":** the age group of the different annotators. Possible values are: 18-22, 23-45, and 46+.
10. **"ethnicity_annotators":** the self-reported ethnicity of the different annotators. Possible values are: "Black or African America", "Hispano or Latino" , "White or Caucasian", "Multiracial", "Asian", "Asian Indian" and "Middle Eastern".
11. **"study_level_annotators":** the self-reported level of study achieved by the different annotators. Possible values are: "Less than high school diploma", "High school degree or equivalent", "Bachelor's degree", "Master's degree" and "Doctorate".
12. **"country_annotators":** the self-reported country where the different annotators live in.

13. **"labels_task2_1":** a set of labels (one for each of the annotators) that indicate if the meme contains sexist expressions or refers to sexist behaviours or not. Possible values are: "YES" and "NO".
14. **"labels_ task2_2":** a set of labels (one for each of the annotators) recording the intention of the person who created the meme. Possible labels are: "DIRECT", "JUDGEMENTAL", "-", and "UNKNOWN".
15. **"labels_ task2_3":** a set of arrays of labels (one array for each of the annotators) indicating the type or types of sexism that are found in the meme. Possible labels are: "IDEOLOGICAL-INEQUALITY","STEREOTYPING-DOMINANCE","OBJECTIFICATION", "SEXUAL-VIOLENCE", "MISOGYNY-NON-SEXUAL-VIOLENCE", "-", and "UNKNOWN".
16. **"split":** subset within the dataset the meme belongs to ("TRAIN-MEME", "TEST-MEME" + "EN"/"ES").

EXIST 2026 Videos Dataset

The EXIST 2026 Videos Dataset contains more than 3,000 labeled Tiktok videos, both in English and Spanish. In particular, the training set contains 2,524 videos and the test set contains 674 videos. Distribution between both languages has been balanced.

The data sets are provided in **JSON format**. Each video is represented as a JSON object with the following attributes:

1. "**id_Tiktok**": a unique identifier in Tiktok platform for the video.
2. "**id_EXIST**": a unique identifier for the video in EXIST evaluation.
3. "**lang**": the languages of the video ("en" or "es").
4. "**text**": the text automatically extracted from the video.
5. "**video**": the name of the file that contains the video.
6. "**path_video**": the path to the file that contains the video.
7. "**url**": the url to the Tiktok video.
8. **"annotators":** a unique identifier for each of the annotators.
9. **"number_annotators":** the number of persons that have annotated the video.
10. **"gender_annotators":** the gender of the different annotators. Possible values are: "F" and "M", for female and male respectively.
11. **"labels_task3_1":** a set of labels (one for each of the annotators) that indicate if the video contains sexist expressions or refers to sexist behaviours or not. Possible values are: "YES", "NO" and "UNKNOWN".
12. **"labels_ task3_2":** a set of labels (one for each of the annotators) recording the intention of the person who created the video. Possible labels are: "DIRECT", "JUDGEMENTAL", "-", and "UNKNOWN".
13. **"labels_ task3_3":** a set of arrays of labels (one array for each of the annotators) indicating the type or types of sexism that are found in the video. Possible labels are: "IDEOLOGICAL-

INEQUALITY","STEREOTYPING-DOMINANCE","OBJECTIFICATION", "SEXUAL-VIOLENCE"**,** "MISOGYNY-NON-SEXUAL-VIOLENCE", "-", and "UNKNOWN".
14. **"split":** subset within the dataset the meme belongs to ("TRAIN-VIDEO", "TEST-VIDEO" + "EN"/"ES").

IMPORTANT: Since labels for subtasks 2.2, 2.3, 3.2 and 3.3 are only assigned if the meme/video has been labeled as YES, the label "-" is assigned to not sexist instances in these subtasks. The label "UNKNOWN" is assigned to memes/videos for which the annotators did not provide a label. **Note that "UNKNOWN" is not a target class for classification**, and therefore should not be predicted by the systems (this labels are discarded in the evaluation phase).

For the test sets, labels for the subtasks are not provided.

Examples of the annotations of a meme and a video from the training sets are given in Figures 1 and 2.


```
"110001": {
  "id_EXIST": "110001",
  "lang": "es",
  "text": "2+2=5 MITO Albert Einstein tenía bajo rendimiento en la escuela. VERDAD 2+2=4 CAN is El feminismo de hoy en día defiende la estupidez humana y no los
  derechos de las mujeres quemo ellas afirman ",
  "meme": "110001.jpeg",
  "path_memes": "memes/110001.jpeg",
  "number_annotators": 6,
  "annotators": ["Annotator_1", "Annotator_2", "Annotator_3", "Annotator_4", "Annotator_5", "Annotator_6"],
  "gender_annotators": ["F", "F", "F", "M", "M", "M"],
  "age_annotators": ["18-22", "23-45", "46+", "46+", "18-22", "23-45"],
  "ethnicities_annotators": ["Hispano or Latino", "Hispano or Latino", "Hispano or Latino", "White or Caucasian", "Hispano or Latino", "Hispano or Latino"],
  "study_levels_annotators": ["High school degree or equivalent", "Master's degree", "Master's degree", "Bachelor's degree", "Bachelor's degree", "Bachelor's
  degree"],
  "countries_annotators": ["Mexico", "Spain", "Argentina", "Spain", "Mexico", "Mexico"],
  "labels_task2_1": ["YES", "YES", "YES", "YES", "YES", "YES"],
  "labels_task2_2": ["DIRECT", "DIRECT", "DIRECT", "DIRECT", "DIRECT", "DIRECT"],
  "labels_task2_3": [
    ["IDEOLOGICAL-INEQUALITY", "STEREOTYPING-DOMINANCE", "MISOGYNY-NON-SEXUAL-VIOLENCE"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"],
    ["IDEOLOGICAL-INEQUALITY"]
  ],
  "split": "TRAIN-MEME_ES"
},
```


Figure 1 Example of the annotations for a meme


```
"120001": {
  "id_Tiktok": 7281385962049998086,
  "id_EXIST": "120001",
  "lang": "es",
  "text": " cuando ves a las del 08 en la fiesta tu amigo el menos salido  j5 [aza red. hay zorro al raimon ha conseguido impedir el tiro de la super táctica.",
  "video": "7281385962049998086.mp4",
  "path_video": "videos/7281385962049998086.mp4",
  "url": "https://www.tiktok.com/@inazumamemes_/video/7281385962049998086",
  "annotators": ["Annotator_4", "Annotator_8", "Annotator_9"],
  "number_annotators": 3,
  "gender_annotators": ["F", "M", "M"],
  "labels_task3_1": ["YES", "NO", "YES"],
  "labels_task3_2": ["DIRECT", "-", "DIRECT"],
  "labels_task3_3": [
    ["OBJECTIFICATION"],
    ["-"],
    ["OBJECTIFICATION"]
  ],
  "split": "TRAIN-VIDEO_ES"
},
```


Figure 2 Example of the annotations for a video

Sensor-Based Experiments and Data Overview

This section presents the physiological ("sensorial") information collected for the multimedia stimuli used in EXIST 2026, including image memes and short video items. These features are intended to complement the stimulus content with measurable signals of attention, arousal, and cognitive processing elicited while subjects viewed each stimulus.

During the experiment[1], subjects were seated comfortably while stimuli were displayed on a screen until they provided a response. A 3-second pause followed each stimulus to minimize carry-over effects between consecutive items. Each session lasted approximately 45 minutes, during which subjects viewed 100–170 stimuli. After each stimulus, they answered brief control questions to ensure engagement and comprehension. All participants provided consent for the anonymous use of their data for research purposes.

Where to Find the Sensorial Data in the JSON

For each instance in the dataset, the physiological data are stored under:

- instance[**"sensorial"**]

And, inside this block, features are organized by subject and modality:

- sensorial[**"users"**]
- sensorial[**"modalities"**][**"ET"**][**"by_user"**]
- sensorial[**"modalities"**][**"HR"**][**"by_user"**]
- sensorial[**"modalities"**][**"EEG"**][**"by_user"**]


```
"countries_annotators": ["Spain", "Portugal", "Spain", "Canada", "United States", "Chile"],
"sensorial": {
  "users": ["ES1", "ES2", "ES3"],
  "modalities": {
    "ET": {
      "by_user": {
        "ES1": {
          "reaction_time": 23014.0,
          "3d_eye_states_pupil diameter left [mm]_mean": 2.7497,
          "3d_eye_states_pupil diameter left [mm]_std": 0.2256,
          "3d_eye_states_pupil diameter left [mm]_max": 4.2079,
          "3d_eye_states_pupil diameter left [mm]_min": 1.7904,
          "3d_eye_states_pupil diameter right [mm]_mean": 2.5381,
          "3d_eye_states_pupil diameter right [mm]_std": 0.1771,
          "3d_eye_states_pupil diameter right [mm]_max": 2.9745,
          "3d_eye_states_pupil diameter right [mm]_min": 1.2602,
          "blinks_duration_mean_ns": 226501504.0
```


Subjects per Stimulus

Each stimulus in the physiological study was viewed by multiple subjects. The **sensorial.users** list

[1] Data collection was approved by the Research Ethics Committee of the Universitat Politècnica de València (Project IDs P01_31-10-2025 and P13_26-12-2025).

contains the identifiers of all subjects associated with a given stimulus across the recorded modalities. In the released dataset, each stimulus has been viewed by **2–4 subjects**.

For each physiological modality—eye tracking (ET), heart rate (HR), and EEG—the corresponding **sensorial.modalities.<MOD>.by_user** dictionary indicates the subset of subjects for which that particular modality was recorded for the stimulus. As a consistency measure, the released files ensure that **each modality includes recordings from at least two subjects** for any stimulus where that modality is available. This guarantees that every stimulus has a minimum amount of data per modality while reflecting the actual number of subjects who participated in each recording.

What is Provided

For each subject and modality, we provide a compact dictionary of summary variables extracted from the physiological signals recorded while the subject viewed the stimulus. Concretely, for a subject identifier like ES1, the feature dictionary is:

- **sensorial.modalities.<MOD>.by_user["ES1"].**

These variables are designed to be lightweight to use: they are aggregated descriptors of the stimulus viewing interval, defined as the time from stimulus onset until the subject advances to the next item by providing their response. The reported values correspond to aggregated statistics over this time window (e.g., mean, standard deviation, minimum, maximum, counts, or durations), rather than instantaneous measurements, and can be fed directly into standard ML pipelines as numerical features.

Modalities and Variables

Eye-Tracking (ET)

ET features summarize how visual attention unfolds while the subject views the stimulus. They can help model which content attracts attention, how strongly it captures gaze, and whether it increases visual/cognitive effort.

- Number of ET variables: 24

Variable groups:

- Pupil diameter (left/right): mean, std, min, max
- Blinks: count and duration statistics (mean, std, min, max)
- Fixations: count and duration statistics (mean, std, min, max)
- Saccades: count and duration statistics (mean, std, min, max)
- Reaction time: time to respond after viewing the stimulus

Why ET can help for sexism-related content: attention and viewing dynamics often change when stimuli contain offensive or emotionally salient cues. Pupil measures are widely used as proxies for cognitive effort and affective load; fixation and saccade patterns reflect how information is sampled; blink behavior can co-vary with attentional demand. Reaction time can reflect processing difficulty or hesitation after exposure to charged content.


```
      "modalities": {
        "ET": {
          "by_user": {
            "ES1": {
              "reaction_time": 23014.0,
              "3d_eye_states_pupil diameter left [mm]_mean": 2.7497,
              "3d_eye_states_pupil diameter left [mm]_std": 0.2256,
              "3d_eye_states_pupil diameter left [mm]_max": 4.2079,
              "3d_eye_states_pupil diameter left [mm]_min": 1.7904,
              "3d_eye_states_pupil diameter right [mm]_mean": 2.5381,
              "3d_eye_states_pupil diameter right [mm]_std": 0.1771,
              "3d_eye_states_pupil diameter right [mm]_max": 2.9745,
              "3d_eye_states_pupil diameter right [mm]_min": 1.2602,
              "blinks_duration_mean_ns": 226501504.0,
              "blinks_duration_std_ns": 75922576.9786,
              "blinks_duration_max_ns": 350375168.0,
              "blinks_duration_min_ns": 145250048.0,
              "blinks_count": 4.0,
              "fixations_duration_mean_ns": 248963299.1549,
              "fixations_duration_std_ns": 220663904.2358,
              "fixations_duration_max_ns": 1351249920.0,
              "fixations_duration_min_ns": 69999104.0,
              "fixations_count": 71.0,
              "saccades_duration_mean_ns": 77578280.2286,
              "saccades_duration_std_ns": 106090482.0967,
              "saccades_duration_max_ns": 555500032.0,
              "saccades_duration_min_ns": 9999872.0,
              "saccades_count": 70.0
            },
            "ES2": {
```


Heart Rate (HR)

HR features summarize autonomic activation during stimulus viewing. They can help capture physiological arousal that may correlate with the emotional impact of the stimulus.

- Number of HR variables: 4

Variables:

- garmin_hr_mean
- garmin_hr_std
- garmin_hr_min
- garmin_hr_max

Why HR can help for sexism-related content: emotionally intense or disturbing material may increase arousal, which can appear as changes in heart-rate level and variability. These descriptors provide a compact signal of autonomic response that can complement purely content-based models.

Electroencephalography (EEG)

EEG features summarize brain activity during stimulus viewing. They can help capture differences in cognitive and affective processing that are not directly observable from the text or visuals alone.

- Number of EEG variables: 80

EEG variables are bandpower features computed for 16 channels (EXG_Channel_0 … EXG_Channel_15), with 5 frequency bands per channel: Delta, Theta, Alpha, Beta, and Gamma bandpower.

Why EEG can help for sexism-related content: different types of stimuli can elicit distinct patterns of engagement, emotional reactivity, and cognitive control. Frequency-domain bandpower features provide compact descriptors of these processes and can support multimodal models that aim to detect or characterize sexist content beyond what is explicitly stated in the stimulus.

Submission Format

The assessment will be conducted utilizing the PyEvALL library (https://github.com/UNEDLENAR/PyEvALL), a specialized Python library for evaluating information systems, therefore submission runs must follow the PyEvALL format according to the evaluation context (a detailed explanation is provided below). Access to the PyEvALL library can be obtained through code implementation or via web by using the online evaluation Framework EvALL (https://evall.uned.es/).

The participants have the possibility to choose:

- The subtask or subtasks they want to participate in.
- For each of the subtasks, whether their system will provide:
    - **Hard labels:** A unique "hard" (single or multiple) label for each instance, as traditionally done.
    - **Soft labels:** A probabilistic distribution over the different possible classes.

Each team is allowed to send up to **3 runs per subtask and type of output (i.e., hard vs soft)**. That is, each team is allowed to send up to **36 runs in total (2x2x9)**. Notice that **runs must executed over the TEST set provided**, runs over dev or training sets will not be accepted.

Submitting Results for subtasks 2.1 and 3.1: Sexism Identification in Tweets/Memes/Videos

Participants submitting results for these subtasks must format the runs in JSON format. Each meme/video must be represented as a JSON object with the following attributes:

1. "**id**": The unique identifier for the meme/video as appear in the dataset ("id_EXIST").
2. **"value":** Possible values are:
   - "YES" or "NO", if a hard output is submitted, or
   - A probability value for each of the two possible labels ("YES" and "NO"), if a soft output is submitted. Note that the sum of the probabilities must be 1.0.
3. **"test_case":** "EXIST2025", as previously indicated, this task builds upon the EXIST 2025 dataset; therefore, for compatibility with previous systems, the evaluation test case name is kept unchanged.

The following images show two examples for two runs (one submitting hard results and the other submitting soft results).

```
[{
  "test_case": "EXIST2025",
  "id": "300001",
  "value": "NO"
}, {
  "test_case": "EXIST2025",
  "id": "300002",
  "value": "NO"
}, {
  "test_case": "EXIST2025",
  "id": "300003",
  "value": "NO"
}, {
  "test_case": "EXIST2025",
  "id": "300004",
  "value": "NO"
}, {
```

(a) Run with hard outputs.

```
[{
  "test_case": "EXIST2025",
  "id": "300001",
  "value": {
    "NO": 0.5,
    "YES": 0.5
  }
}, {
  "test_case": "EXIST2025",
  "id": "300002",
  "value": {
    "YES": 0.8333333333333334,
    "NO": 0.16666666666666666
  }
}, {
  "test_case": "EXIST2025",
  "id": "300003",
  "value": {
    "NO": 1.0,
    "YES": 0.0
  }
}, {
```

(b) Run with soft outputs

Note that, if desired, the participants can provide only hard or soft labels, or both labels (in separated files).

Submitting Results for subtask 2.2 and 3.2: Source Intention in Memes/Videos

Participants submitting results for these subtasks must format the runs in JSON format. Each meme/video must be represented as a JSON object with the following attributes:

1. "**id**": The unique identifier for the meme/video as appear in the dataset ("id_EXIST").
2. **"value":** Possible values are:

- "NO", "DIRECT" or "JUDGEMENTAL".
- A probability value for each of the 3 possible labels, if a soft output is submitted. Note that the sum of the probabilities must be 1.0.

3. **"test_case":** "EXIST2025", as previously indicated, this task builds upon the EXIST 2025 dataset; therefore, for compatibility with previous systems, the evaluation test case name is kept unchanged.

The following images show two examples for two runs (one submitting hard results and the other submitting soft results).

```
[{
  "test_case": "EXIST2025",
  "id": "300002",
  "value": "JUDGEMENTAL"
}, {
  "test_case": "EXIST2025",
  "id": "300003",
  "value": "NO"
}, {
  "test_case": "EXIST2025",
  "id": "300004",
  "value": "REPORTED"
}, {
```

(a) Run with hard outputs.

```
[{
  "test_case": "EXIST2025",
  "id": "110001",
  "value": {
    "DIRECT": 1.0,
    "NO": 0.0,
    "JUDGEMENTAL": 0.0
  }
}, {
  "test_case": "EXIST2025",
  "id": "110002",
  "value": {
    "DIRECT": 0.8333333333333334,
    "JUDGEMENTAL": 0.16666666666666666,
    "NO": 0.0
  }
}, {
```

(b) Run with soft outputs

Again, the participants can provide only hard or soft labels, or both labels (in separated files).

<u>Submitting Results for subtasks 2.3 and 3.3: Sexism Categorization in Memes/Videos</u>

Participants submitting results for these subtasks must format the runs in JSON format. Each meme/video must be represented as a JSON object with the following attributes:

1. "**id**": The unique identifier for the tweet/meme/video as appear in the dataset ("id_EXIST").
2. **"hard_label":** Possible values are:
   - "NO", ""IDEOLOGICAL-INEQUALITY", "STEREOTYPING-DOMINANCE", "OBJECTIFICATION", "SEXUAL-VIOLENCE" and "MISOGYNY-NON-SEXUAL-VIOLENCE".
   - A probability value for each of the six possible labels, if a soft output is submitted. Note that, since this is a multi-label classification task, the sum of the probabilities does not have to be 1.0.
3. **"test_case":** "EXIST2025", as previously indicated, this task builds upon the EXIST 2025 dataset; therefore, for compatibility with previous systems, the evaluation test case name is kept unchanged.

The following images show two examples for two runs (one submitting hard results and the other submitting soft results).


```
[{
  "test_case": "EXIST2025",
  "id": "300002",
  "value": ["IDEOLOGICAL-INEQUALITY", "MISOGYNY-NON-SEXUAL-VIOLENCE"]
}, {
  "test_case": "EXIST2025",
  "id": "300003",
  "value": ["NO"]
}, {
  "test_case": "EXIST2025",
  "id": "300004",
  "value": ["SEXUAL-VIOLENCE"]
}, {
```


(a) Run with hard outputs.


```
[{
  "test_case": "EXIST2025",
  "id": "300001",
  "value": {
    "NO": 0.5,
    "MISOGYNY-NON-SEXUAL-VIOLENCE": 0.5,
    "SEXUAL-VIOLENCE": 0.16666666666666666,
    "IDEOLOGICAL-INEQUALITY": 0.0,
    "STEREOTYPING-DOMINANCE": 0.0,
    "OBJECTIFICATION": 0.0
  }
}, {
  "test_case": "EXIST2025",
  "id": "300002",
  "value": {
    "IDEOLOGICAL-INEQUALITY": 0.3333333333333333,
    "STEREOTYPING-DOMINANCE": 0.3333333333333333,
    "MISOGYNY-NON-SEXUAL-VIOLENCE": 0.5,
    "NO": 0.16666666666666666,
    "OBJECTIFICATION": 0.16666666666666666,
    "SEXUAL-VIOLENCE": 0.0
  }
}, {
```


(b) Run with soft outputs.

Again, the participants can provide only hard or soft labels, or both labels (in separated files).

How to Submit your Runs

Each team must pack all the runs in a directory named

```
exist2026_<team_name>
```

The directory will contain one file per run named

```
<task>_<subtask>_<evaluation_context>_<team_name>_<run_id>
```

where **run_id** is a number between 1 and 3, **evaluation context** can be *hard* or *soft*, **task** may be *task2* or *task3*, and **subtask** may be *1, 2* or *3*.

For instance:
- exist2026_UNED/task2_2_hard_UNED_1
- exist2026_UNED/task3_3_soft_UNED_3

The (compressed) directory with your runs must be submitted to the competition by filling up the following form: https://forms.gle/5hY91c7aBv563oZM7.
Notice that only one submission per team is allowed.

Evaluation

From the point of view of evaluation metrics, our six subtasks can be described as:

- **Subtasks 2.1 and 3.1 (sexism identification)**: binary classification, mono label.
- **Subtasks 2.2 and 3.2 (source intention)**: multiclass hierarchical classification, mono label. The hierarchy of classes has a first level with YES/NO, and a second level for the sexist category with two mutually exclusive subcategories: *direct* and *judgemental*. A suitable evaluation metric must reflect the fact that a confusion between not sexist and a sexist category is more severe than a confusion between two sexist subcategories.
- **Subtasks 2.3 and 3.3 (sexism categorization)**: multiclass hierarchical classification, multi-label. Again, the first level is a binary distinction between YES/NO, and there is a second level for the sexist category that includes "*ideological and inequality", "stereotyping and dominance", "objectification", "sexual violence"* and *"misogyny and non-sexual violence"*. These classes are not mutually exclusive: a tweet/meme/video may belong to several subcategories at the same time.

The learning with disagreements paradigm can be considered in both sides of the evaluation process:

(i) **The ground truth**. In a "hard" setting, variability in the human annotations is reduced to a gold standard set of categories, **hard labels,** that are assigned to each item (e.g., using majority vote). In a "soft" setting, the gold standard is the full set of human annotations with their variability. Therefore, the evaluation metric incorporates the proportion of human annotators that have selected each category, **soft labels**. Note that in subtasks 2.1, 3.1, 2.2 and 3.2, which are mono label problems, the sum of the probabilities of each class must be one. But in subtasks 2.3 and 3.3, which

are multi-label, each annotator may select more than one category for a single item. Therefore, the sum of the probabilities of each class may be larger than one.
(ii) **The system output**. In a “hard”, traditional setting, the system predicts one or more categories for each item. In a “soft” setting, the system predicts a probability for each category, for each item. The evaluation score is maximized when the probabilities predicted match the actual probabilities in a soft ground truth. Again, note that in subtasks 2.3 and 3.3, which is a multi-label problem, the probabilities predicted by the system for each of the categories do not necessarily add up to one.

For each of the tasks, two types of evaluation will be reported:

1. ***Hard-hard*:** hard system output and hard ground truth.
2. ***Soft-soft*:** soft system output and soft ground truth.

OFFICIAL METRIC: ICM & ICM-soft

For all tasks and all types of evaluation (hard-hard, hard-soft and soft-soft) we will use the same official metric: ICM (Information Contrast Measure) (Amigó and Delgado, 2022). ICM is a similarity function that generalizes Pointwise Mutual Information (PMI), and can be used to evaluate system outputs in classification problems by computing their similarity to the ground truth categories. The general definition of ICM is:

$$\mathrm{ICM}(A,B) = \alpha_1 IC(A) + \alpha_2 IC(B) - \beta IC(A \cup B)$$

Where IC(A) is the Information Content of the item represented by the set of features A, etc. ICM maps into PMI when all parameters take a value of 1.

In Amigó and Delgado, the general ICM definition is applied to cases where categories have a hierarchical structure and items may belong to more than one category. The resulting evaluation metric is proved to be analytically superior to the alternatives in the state of the art. The definition of ICM in this context is:

$$\mathrm{ICM}(s(d), g(d)) = 2I(s(d)) + 2I(g(d)) - 3I(s(d) \cup g(d))$$

Where I() stands for Information Content, s(d) is the set of categories assigned to document d by system s, and g(d) the set of categories assigned to document d in the gold standard.

As there is not, to the best of our knowledge, any current metric that fits hierarchical multi-label classification problems in a learning with disagreement scenario, we have defined an extension of ICM (*ICM-soft*) that accepts both soft system outputs and soft ground truth assignments. ICM-soft works as follows: first, we define the Information Content of a single assignment of a category c with an agreement v to a given item:

$$I(\{\langle c, v\rangle\}) = -log_2(P(\{d \in D : g_c(d) \geq v\})$$

Note that the information content of assigning a category c with an agreement v grows inversely with the probability of finding an item that receives category c with agreement equal or larger than v.

The system output and the gold standards are sets of assignments. Therefore, in order to estimate their information content, we apply a recursive function similar to the one described in (Amigó and Delgado, 2022):

$$I\left(\bigcup_{i=1}^{n}\{\langle c_i, v_i\rangle\}\right) = I(\langle c_1, v_1\rangle) + I\left(\bigcup_{i=2}^{n}\{\langle c_i, v_i\rangle\}\right)$$
$$- I\left(\bigcup_{i=2}^{n}\{\langle \texttt{lca}(c_1, c_i), min(v_1, v_i)\rangle\}\right)$$

where lca(a,b) is the lowest common ancestor of categories a and b.

EVALUATION VARIANTS FOR EACH TASK

For each of the tasks, the evaluation will be performed in the two modes described above, as follows:

- **Hard-hard evaluation**. For systems that provide a hard, conventional output, we will provide a hard-hard evaluation. To derive the hard labels in the ground truth from the different annotators' labels, we use a probabilistic threshold computed for each task. As a result, for 2.1, the class annotated by more than 3 annotators is selected; for subtask 2.2, the class annotated by more than 2 annotators is selected; and for subtask 2.3 (multi-label), the class annotated by more than 1 annotator are selected. Due to the nature of subtasks 3.1, 3.2 and 3.3 and the complexity of video labeling, the labelling methodology challenged for this subtasks so hard labels included are those annotated by more than 1 annotator. Items for which there is no majority class (i.e., no class receives more probability than the threshold) will be removed from this evaluation scheme. **The official metric will be the original ICM** (as defined in (Amigó and Delgado, 2022)). We will also report and compare systems with F1 (the harmonic average of precision and recall). In subtasks 2.1 and 3.1, we will use F1 for the positive class. In the remaining subtasks, we will use the average of F1 for all classes. Note, however, that F1 is not ideal in our experimental setting: although it can handle multi-label situations, it does not consider the relationships between classes: a mistake between not sexist and any of the sexist subclasses, and a mistake between two of the positive subclasses, are penalized equally, although the former is a more severe error.
- **Soft-soft evaluation**. For systems that provide probabilities for each category, we will provide a soft-soft evaluation that compares the probabilities assigned by the system with

the probabilities assigned by the set of human annotators. As in the previous case, **we will use ICM-soft as the official evaluation metric in this variant**. We may also report additional metrics in the final report.

Description of the PyEvALL Evaluation Library

The EXIST evaluation campaigns use the PyEvALL evaluation library, which includes all the metrics used in the EXIST 2026 competition. PyEvALL is available at https://github.com/UNEDLENAR/PyEvALL. PyEvALL (The Python library to Evaluate ALL) is an evaluation tool for information systems that allows assessing a wide range of metrics covering various evaluation contexts, including classification, ranking, or LeWiDi (Learning with disagreement). The documentation included in the PyEvALL Readme explains how to install it (via pip or source code), how to use it, and the input and output formats (the different PyEvALL reports).

How to use the official PyEvALL python evaluation package

PyEvALL allows to evaluate several metrics from different evaluation contexts using only 8 lines of code. As an example, the following code will execute an evaluation over a prediction file for the EXIST subtask 2.1, and will show the results in an embedded report:

```
predictions = "test/hard/EXIST2025_test_task2_1_baseline_2.json"
gold = "test/hard/EXIST2025_test_task2_1_gold_hard.json"
test = PyEvALLEvaluation()
params= dict()
params[PyEvALLUtils.PARAM_REPORT]= PyEvALLUtils.PARAM_OPTION_REPORT_EMBEDDED
metrics=["ICM", "ICMNorm" ,"FMeasure"]
report= test.evaluate(predictions, gold, metrics, **params)
report.print_report()
```

Similarly, PyEvALL allows to evaluate several prediction files and generate a meta report. As an example, the following code will execute an evaluation over different prediction files for the EXIST subtask 2.2, and will show the results in a table in tsv format:

```
predictions = "test/hard/"
lst_pred=[]
onlyfiles = [f for f in listdir(predictions) if isfile(join(predictions, f))]
for file in onlyfiles:
     lst_pred.append(predictions+ file)

gold = "test/hard/EXIST2025_test_task2_2_gold_hard.json"
TASK2_2_HIERARCHY = {"YES":["DIRECT","JUDGEMENTAL"], "NO":[]}
params= dict()
prueba = PyEvALLEvaluation()
params[PyEvALLUtils.PARAM_HIERARCHY]= TASK2_2_HIERARCHY
params[PyEvALLUtils.PARAM_REPORT]= PyEvALLUtils.PARAM_OPTION_REPORT_DATAFRAME
metrics=["ICM", "ICMNorm" ,"FMeasure"]
report= prueba.evaluate_lst(lst_pred, gold, metrics, **params)
report.print_report_tsv()
```

Notice that, as subtask 1.2 is a hierarchical classification problem, the parameter PyEvALLUtils.PARAM_HIERARCHY has been set with the appropriate hierarchy.

The evaluation of all subtasks in the EXIST 2026 challenge can be done using the code previously described by changing the configuration, metrics and hierarchy, according to the different context evaluations:

Hard-hard subtasks 2.1 and 3.1: in this configuration, there is no hierarchy and the measures used are ICM, ICM Normalized and F1.

```
metrics=["ICM", "ICMNorm" ,"FMeasure"]
```

Hard-hard subtasks 2.2 and 3.2: in this configuration, the hierarchy parameter should be set as well as the measures ICM, ICM Normalized and F1.

```
metrics=["ICM", "ICMNorm" ,"FMeasure"]
TASK2_2_HIERARCHY = {"YES":["DIRECT, "JUDGEMENTAL"], "NO":[]}
params[PyEvALLUtils.PARAM_HIERARCHY]= TASK2_2_HIERARCHY
```

Hard-hard subtasks 2.3 and 3.3: in this configuration, the hierarchy parameter should be set as well as the measures ICM, ICM Normalized and F1.

```
metrics=["ICM", "ICMNorm" ,"FMeasure"]
TASK2_3_HIERARCHY = {"YES":["IDEOLOGICAL-INEQUALITY","STEREOTYPING-
DOMINANCE","OBJECTIFICATION", "SEXUAL-VIOLENCE", "MISOGYNY-NON-SEXUAL-
VIOLENCE"], "NO":[]}
params[PyEvALLUtils.PARAM_HIERARCHY]= TASK2_3_HIERARCHY
```

Soft-soft subtasks 2.1 and 3.1: in this configuration, there is no hierarchy and the measures used are ICM-soft, ICM-soft Normalized and F1.

```
metrics=["ICMSoft", "ICMSoftNorm", "CrossEntropy"]
```

Soft-soft subtasks 2.2 and 3.2: in this configuration the hierarchy parameter should be set as well as the measures ICM-soft, ICM-soft Normalized and F1.

```
metrics=["ICMSoft", "ICMSoftNorm", "CrossEntropy"]
TASK2_2_HIERARCHY = {"YES":["DIRECT, "JUDGEMENTAL"], "NO":[]}
params[PyEvALLUtils.PARAM_HIERARCHY]= TASK2_2_HIERARCHY
```

Soft-soft subtasks 2.3 and 3.3: in this configuration the hierarchy parameter should be set as well as the measures ICM-soft, ICM-soft Normalized and F1.

```
metrics=["ICMSoft", "ICMSoftNorm"]
TASK2_3_HIERARCHY = {"YES":["IDEOLOGICAL-INEQUALITY","STEREOTYPING-
DOMINANCE","OBJECTIFICATION", "SEXUAL-VIOLENCE", "MISOGYNY-NON-SEXUAL-
VIOLENCE"], "NO":[]}
params[PyEvALLUtils.PARAM_HIERARCHY]= TASK2_3_HIERARCHY
```

Content of the evaluation folder

The content of the folder evaluation is:

- **“golds”**: this folder includes the official gold standards for training in all tasks and evaluation contexts. In particular, the “hard_gold” allows participants to evaluate their system outputs in a hard-hard evaluation context, while the “soft_gold” must be used in the soft-soft evaluations.
- **“baselines":** this folder includes the official baselines for each task. Notice that, with only one file, all evaluation contexts can be addressed. The “majority_class” baseline is basically a non-informative system where all instances are labeled with the majority class, while the “minority_class” is a non-informative system where all instances are classified as the minority class. Notice that these baselines are provided only as an example of system output, and not as a state-of-the-art approximation.

Questions

If you have any questions or problems, please open a thread on the support-forum on Discord.

References

Overview of Previous LeWiDi EXIST Editions:

- Laura Plaza, Jorge Carrillo-De-Albornoz, Iván Arcos, Paolo Rosso, Damiano Spina, Enrique Amigó, Julio Gonzalo, Roser Morante. **Overview of EXIST 2025: Learning with Disagreement for Sexism Identification and Characterization in Tweets, Memes, and TikTok Video.** Carrillo-De-Albornoz, J., et al. Experimental IR Meets Multilinguality, Multimodality, and Interaction. CLEF 2025. Lecture Notes in Computer Science, vol 16089. Springer, Cham. https://link.springer.com/book/10.1007/978-3-032-04354-2.

- Laura Plaza, Jorge Carrillo-de-Albornoz, Victor Ruiz, Alba Maeso, Berta Chulvi, Paolo Rosso, Enrique Amigó, Julio Gonzalo, Roser Morante and Damiano Spina. (2024). **Overview of EXIST 2024 - Learning with Disagreement for Sexism Identification and Characterization in Tweets and Memes**. In L. Goeuriot et al. (Eds.), Experimental IR Meets Multilinguality, Multimodality, and Interaction (pp. 93–117). Cham: Springer Nature Switzerland.

- Laura Plaza, Jorge Carrillo-de-Albornoz, Roser Morante, Enrique Amigó, Julio Gonzalo, Damiano Spina & Paolo Rosso . **Overview of EXIST 2023 – Learning with Disagreement for Sexism Identification and Characterization.** Arampatzis, A., et al. Experimental IR Meets Multilinguality, Multimodality, and Interaction. CLEF 2023. Lecture Notes in Computer Science, vol 14163. Springer, Cham. https://doi.org/10.1007/978-3-031-42448-9_23.

Extended Overview of Previous LeWiDi EXIST Editions:

- Laura Plaza, Jorge Carrillo-De-Albornoz, Iván Arcos, Paolo Rosso, Damiano Spina, Enrique Amigó, Julio Gonzalo, Roser Morante. **Overview of EXIST 2025: Learning with Disagreement for Sexism Identification and Characterization in Tweets, Memes, and TikTok Videos (Extended Overview).** Guglielmo Faggioli et al.Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2025). CEUR Working Notes Vol-4038.

- Laura Plaza, Jorge Carrillo de Albornoz, Víctor Ruiz, Alba Maeso, Berta Chulvi, Paolo Rosso, Enrique Amigó, Julio Gonzalo, Roser Morante, Damiano Spina: **Overview of EXIST 2024 - Learning with Disagreement for Sexism Identification and Characterization in Tweets and Memes (Extended Overview)**. Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2024). CEUR Working Notes Vol-3740.: 908-941

- Laura Plaza, Jorge Carrillo-de-Albornoz, Roser Morante, Enrique Amigó, Julio Gonzalo, Damiano Spina & Paolo Rosso . **Overview of EXIST 2023 – Learning with Disagreement for Sexism Identification and Characterization (Extended Overview).** Mohammad Aliannejadi, et al.Working Notes of the Conference and Labs of the Evaluation Forum (CLEF 2023). CEUR Working Notes Vol-3497.

Working Notes of Previous LeWiDi EXIST Editions:

- All 2025 proceedings available at: EXIST 2025 Proceedings.
- All 2024 proceedings available at: EXIST 2024 Proceedings.
- All 2023 proceedings available at: EXIST 2023 Proceedings.

Video and Meme Related Work

- Iván Arcós and Paolo Rosso (2024) **Sexism Identification on TikTok: A Multimodal AI Approach with Text, Audio, and Video**. In: Proc. 15th Int. Conf. of the CLEF Association, CLEF 2024, Grenoble, France, September 9-12, Springer-Verlag, LNCS(14958), pp. 61-73

- Italiani, P., Gimeno-Gómez, D., Ragazzi, L., Moro, G., Rosso, P. (2026). MEMEWEAVER: Inter-Meme Graph Reasoning for Sexism and Misogyny Detection. In Findings EACL 2026.

- Arcos, I., Rosso, P., Gomis, E. (2026). Human-Centered Multimodal Fusion for Sexism Detection in Memes with Eye-Tracking, Heart Rate, and EEG Signals. In Proceedings LREC 2026 http://arxiv.org/abs/2602.23862

Sensor Data and NLP Related Work

- Alaçam, Ö., Hoeken, S., & Zarrieß, S. (2024). **Eyes Don't Lie: Subjective Hate Annotation and Detection with Gaze**. In Proceedings of EMNLP 2024 (pp. 187–205).

- Zhang, Y., Li, Q., Nahata, S., Jamal, T., Cheng, S., Cauwenberghs, G., & Jung, T.-P. (2024). **Integrating LLMs, EEG, and eye-tracking for word-level neural state classification**. IEEE Transactions on Neural Systems and Rehabilitation Engineering, 32, 3465–3475.

- Oğuz, F. E., Alkan, A., & Schöler, T. (2023). **Emotion detection from ECG signals with different learning algorithms and automated feature engineering**. Signal, Image and Video Processing, 17(7), 3783–3791.

- Yoon, H., Tolera, B. A., Gong, T., Lee, K., & Lee, S.-J. (2024). **By My Eyes: Grounding multimodal large language models with sensor data via visual prompting**. In Proceedings of EMNLP 2024 (pp. 2219–2241).

- Kaixin Ji, Danula Hettiachchi, Falk Scholer, Flora D. Salim, and Damiano Spina. 2025. SenseSeek **Dataset: Multimodal Sensing to Study Information Seeking Behaviors.** Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 9, 3, Article 92 (September 2025), 29 pages. https://doi.org/10.1145/3749501